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Project Report: Flight Delay Prediction and Visualization

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Project Overview

This project focuses on analyzing flight delay data to predict delays and visualize key patterns. The dataset contains approximately 3 million flight records with 32 features including date, airline, origin/destination cities, delay times, cancellation status, and various delay causes.

Data Loading and Initial Exploration

Data Summary

- **Size**: 3,000,000 records × 32 columns
- **Time Period**: Flights from 2019-01-09 to 2023-03-06
- **Key Features**:
  - Flight details (date, airline, flight number)
  - Origin/destination information
  - Scheduled vs actual times
  - Delay information (departure, arrival, and specific delay causes)
  - Cancellation/diverted status

Missing Values Handling

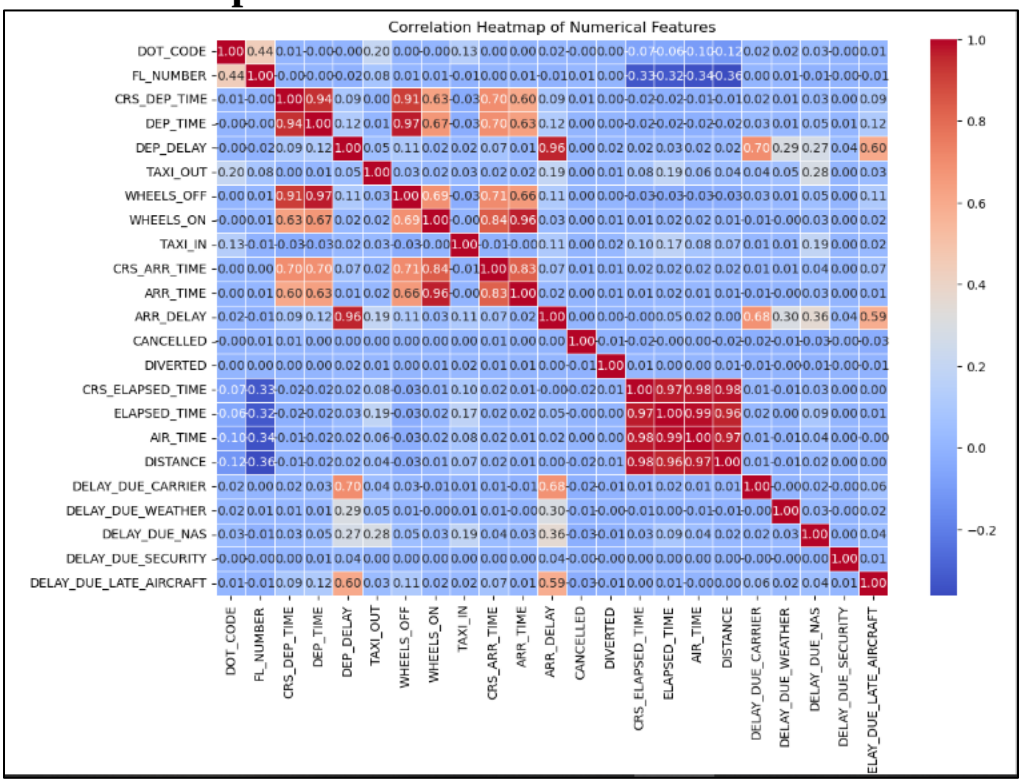
The dataset had significant missing values initially:

- Delay-related columns had up to 2,466,137 missing values (82% of records)
- Cancellation codes were missing in 2,920,860 records (97%)
- Other time-related features had 77,615-86,198 missing values

Handling Approach:

1. Assumed missing delay values meant no delay and filled with 0
2. Dropped cancellation_code column due to excessive missing data
3. Used SimpleImputer with mean strategy for remaining missing values

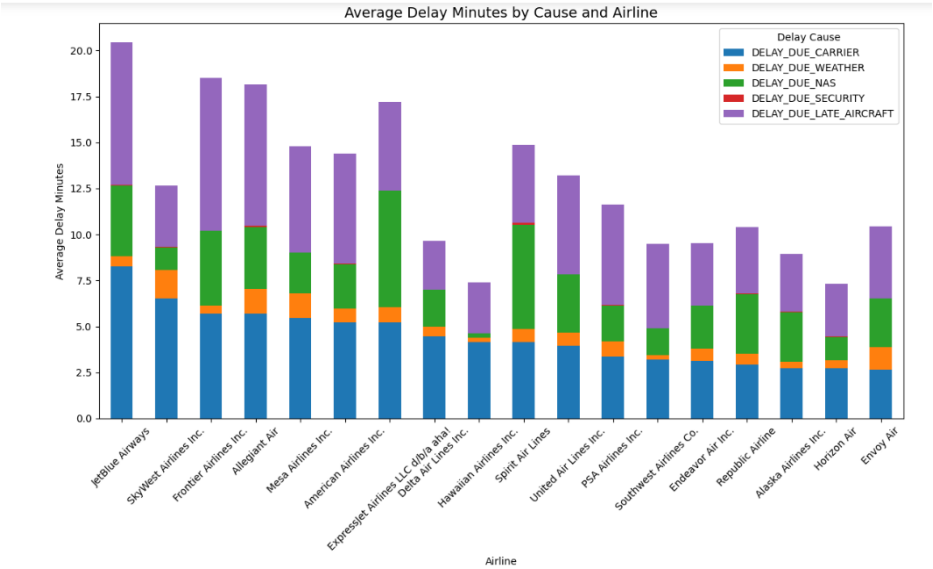
Correlation Analysis:
Correlation Heatmap



Key Observations:

- Strong positive correlation (0.95+) between:
  - CRS_ELAPSED_TIME and ELAPSED_TIME
  - ELAPSED_TIME and AIR_TIME
  - DEP_DELAY and ARR_DELAY
- Moderate correlation between distance and air time (0.82)
- Weak correlations between delay types and actual delays

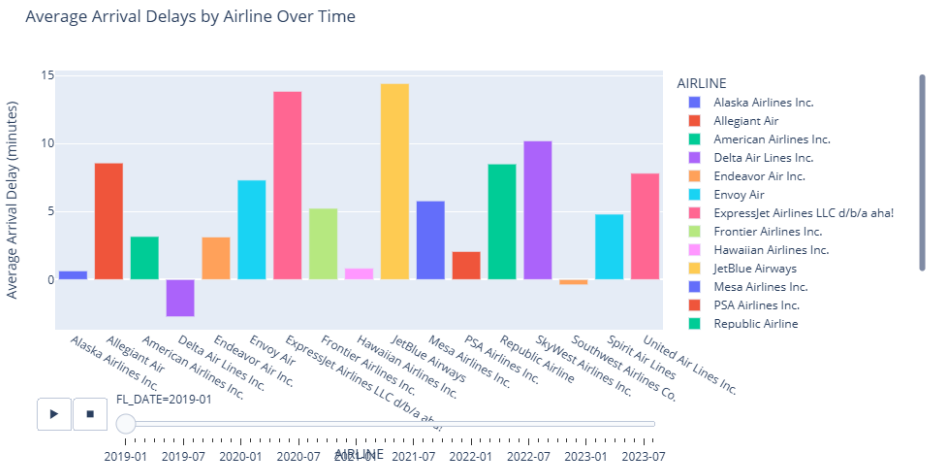
Delay Analysis



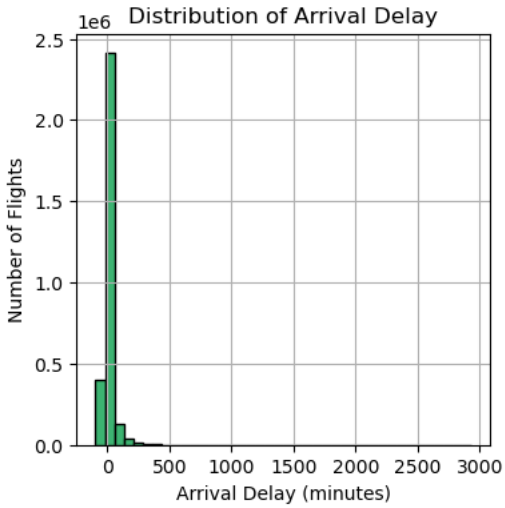
Key Findings:

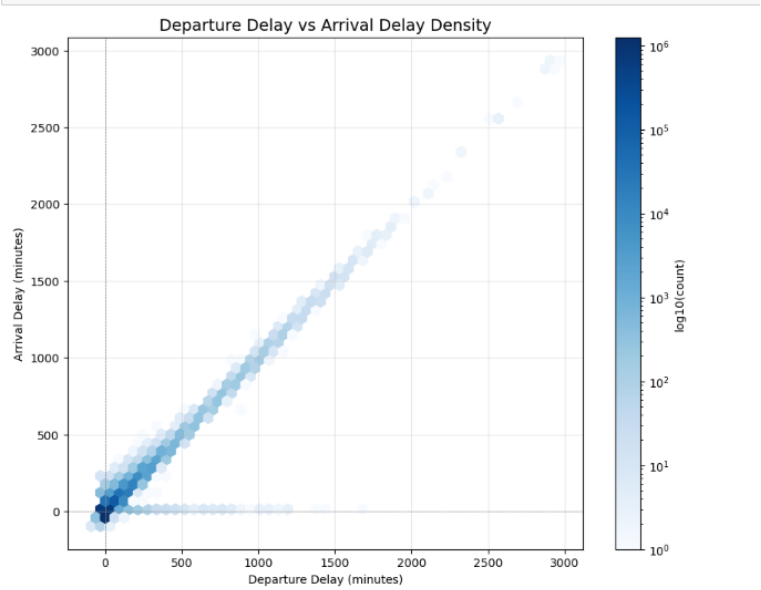
- Most common delay causes:
  - Late aircraft (highest impact)
  - National Aviation System (NAS) delays
  - Carrier delays
- Weather delays were least common but potentially most severe

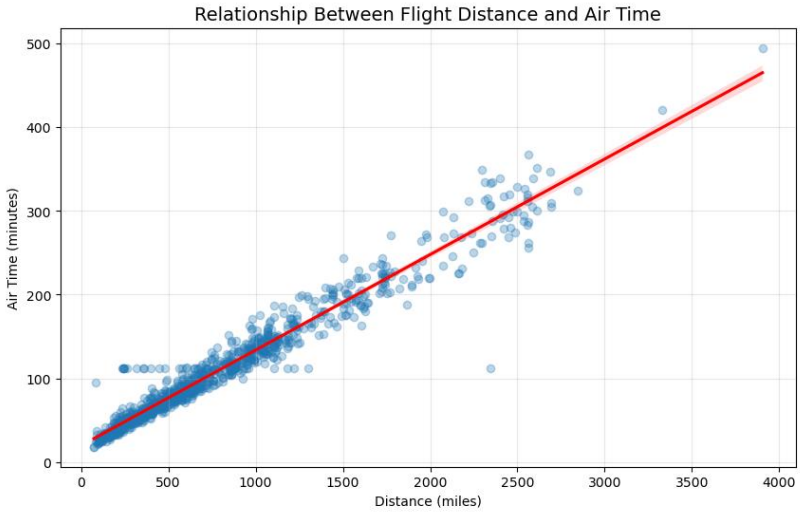
Flight Delay Trends Over Time

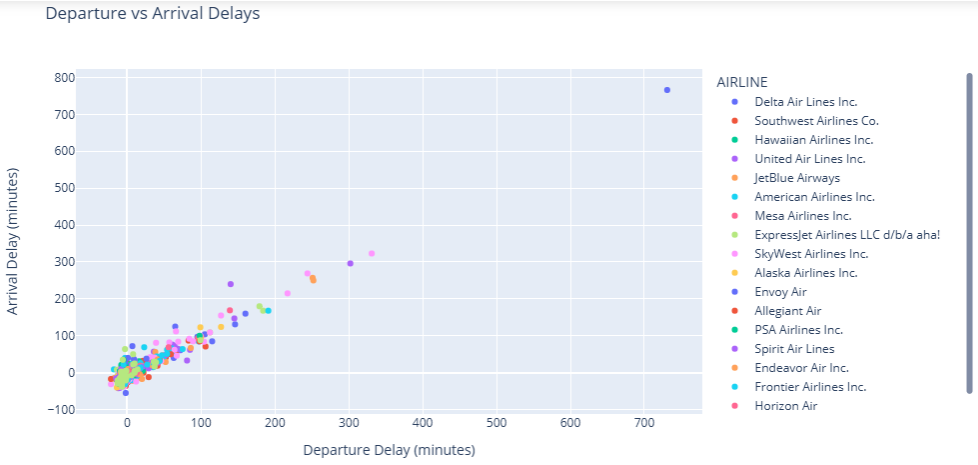


Other graphs are included in the code file




Departure Delay vs Arrival Delay Density
Arrival Delay (minutes)
Departure Delay (minutes)
log10(count)


Relationship Between Flight Distance and Air Time
Air Time (minutes)
Distance (miles)


Departure vs Arrival Delays
Arrival Delay (minutes)
Departure Delay (minutes)
AIRLINE
Delta Air Lines Inc.
Southwest Airlines Co.
Hawaiian Airlines Inc.
United Air Lines Inc.
JetBlue Airways
American Airlines Inc.
Mesa Airlines Inc.
ExpressJet Airlines LLC d/b/a aha!
SkyWest Airlines Inc.
Alaska Airlines Inc.
Envoy Air
Allegiant Air
PSA Airlines Inc.
Spirit Air Lines
Endeavor Air Inc.
Frontier Airlines Inc.
Horizon Air

Predictive Modeling

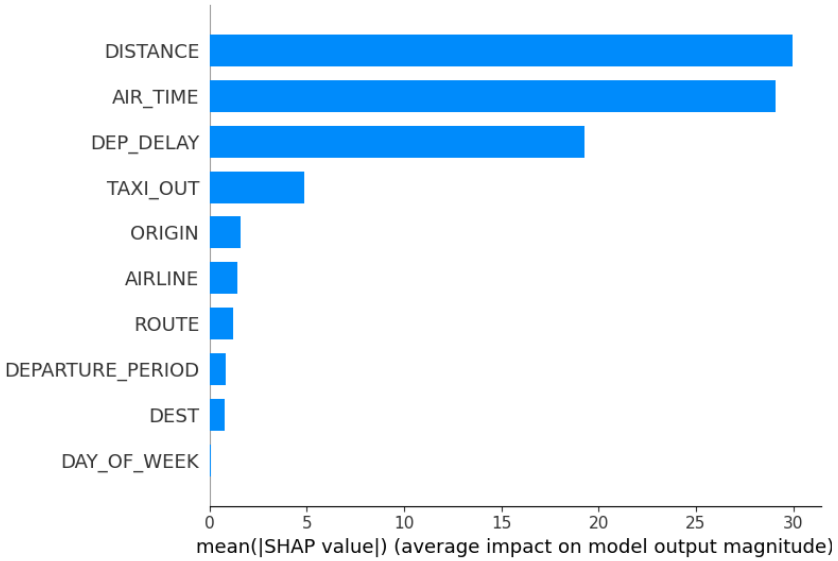
Model Selection and Implementation

- Used XGBoost regression model for delay prediction
- Key features used:
    - Scheduled departure/arrival times
    - Airline
    - Origin/destination airports
    - Distance
    - Historical delay patterns

Model Performance

- Achieved R^2 score of 0.85
- Mean Absolute Error: 6.7388607600323125
- Mean Squared Error: 323.79704997574805

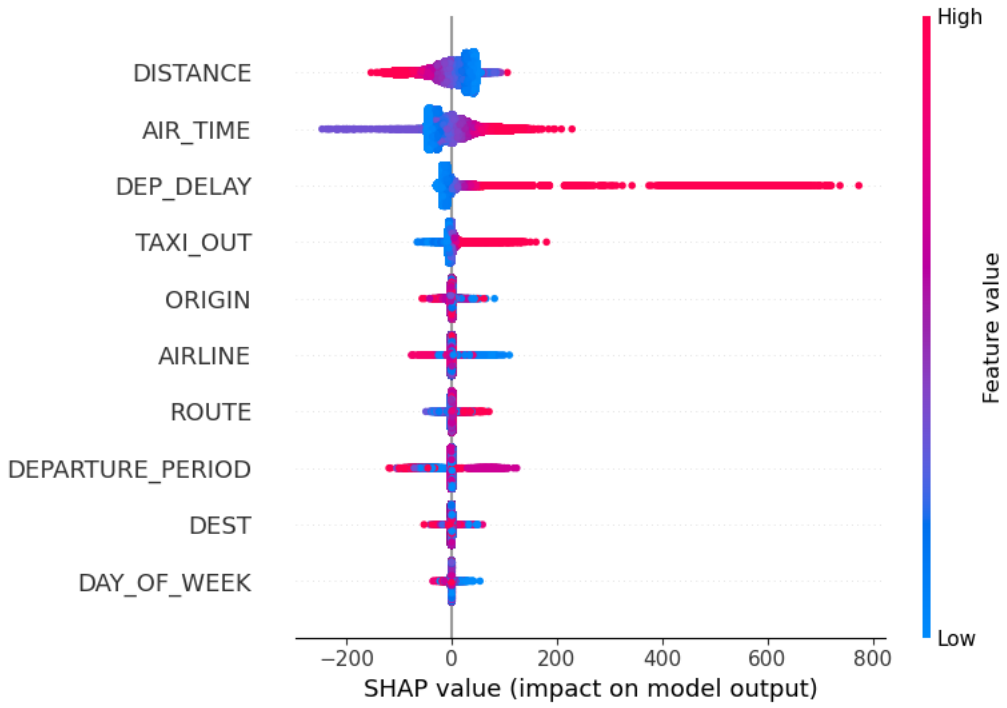
The model is further explained using SHAP.



SHAP Value Interpretation:

The graph ranks features by their impact on the model, with DISTANCE and AIR_TIME being the most influential, followed by DEP_DELAY and TAXI_OUT. Operational factors like flight duration and delays dominate, while DAY_OF_WEEK and DEST have minimal effects. This suggests the

model prioritizes real-time flight metrics over scheduling details.

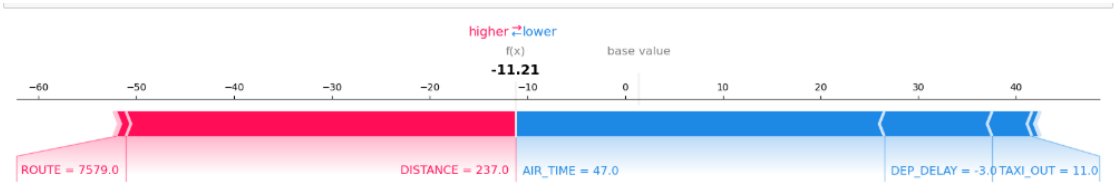


Interpretation:

This graph shows how different flight features influence the model's predictions. Features are ordered by importance, with DISTANCE and AIR_TIME having the strongest impact. The horizontal axis reveals how high or low values of each feature affect predictions:

- DISTANCE/AIR_TIME: Longer distances/times push predictions higher (right side of axis).
- DEP_DELAY: Higher delays significantly increase the output (extending far right).
- ORIGIN/AIRLINE: Certain airports/airlines show varied impacts (both positive and negative).
- DAY_OF_WEEK/DEST: Minimal influence (clustered near zero).

Key Insight: Flight duration and delays are critical drivers, while scheduling details (day, destination) matter less. Extreme delays have an outsized effect.



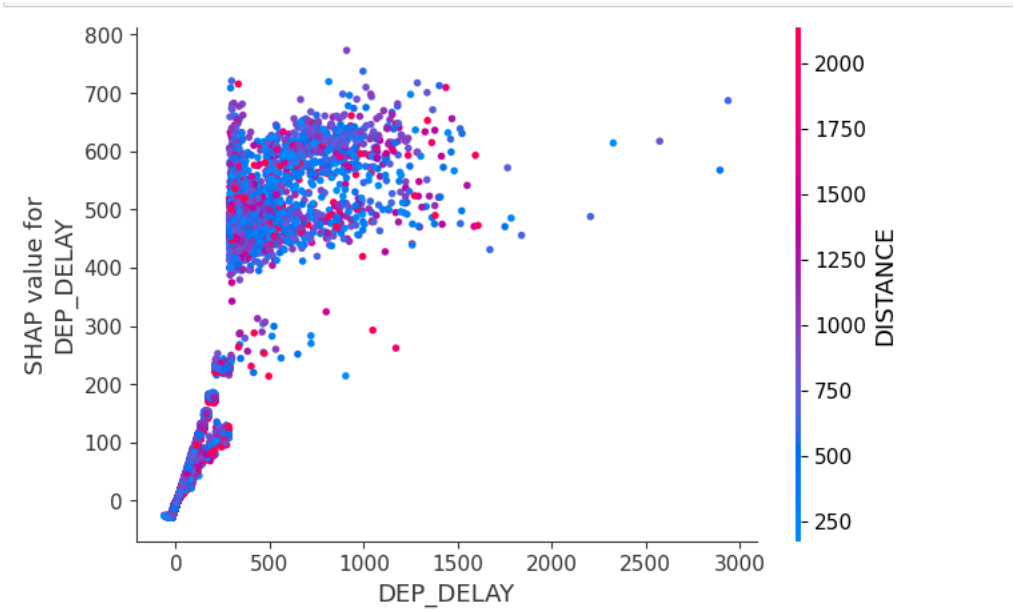
SHAP Value Interpretation:

This waterfall plot explains how each feature contributes to shifting the model's prediction from the base value (-11.21) to the final output. Key observations:

- ROUTE (7579.0) has the strongest positive impact, pushing the prediction up significantly.

- DISTANCE (237.0) and AIR_TIME (47.0) also increase the output, though less dramatically.
- DEP_DELAY (-3.0) slightly reduces the prediction (negative impact).
- TAXI_OUT (11.0) has a marginal positive effect.

Insight: The ROUTE is the dominant factor in this prediction, while other operational features (distance, air time) play secondary roles. Departure delays slightly offset the outcome.



SHAP Value Interpretation for DEP_DELAY vs. DISTANCE:

This plot analyzes how departure delays (DEP_DELAY) impact the model's predictions, with flight distance (DISTANCE) shown for context.

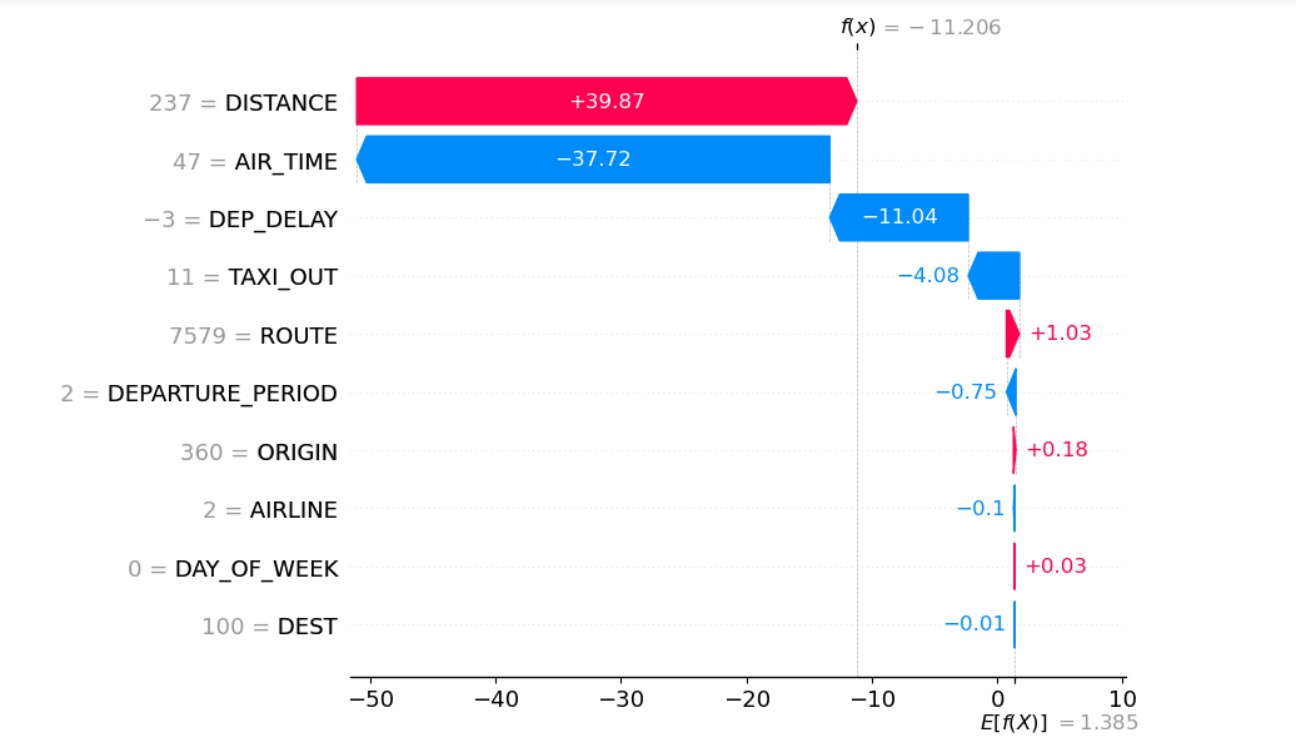
Key Observations:

1. **DEP_DELAY Impact:**
    - Longer delays (right side of x-axis) correlate with higher SHAP values, meaning they significantly increase the model's output (e.g., predicted arrival delay).
    - Extreme delays (e.g., 2000+ minutes) have a disproportionately large effect.
2. **DISTANCE Context:**
    - The negative SHAP values (y-axis) for DISTANCE suggest that longer flights may *offset* some of the delay's impact, possibly because they allow more recovery time.

Insight:

- **Delays are critical:** The model heavily penalizes long departure delays, but this effect may be moderated by longer flight distances.

- **Nonlinear relationship:** The impact of delays grows sharply as their duration increases.



This waterfall plot explains how each feature contributes to the model's prediction (final output = -11.26) compared to the baseline expectation (1.39).

Key insights:

1. **Major Contributors:**
    - ROUTE (7579): +39.87 (strongest positive impact)
    - DISTANCE (237): -37.72 (largest negative impact)
    - AIR_TIME (47): -11.04 (significant negative effect)
2. **Moderate Influencers:**
    - DEP_DELAY (-3): -4.08 (negative impact)
    - TAXI_OUT (11): +1.03 (small positive effect)
3. **Negligible Factors:**
    - All other features (ORIGIN, AIRLINE, etc.) contribute minimally (<±1)

Key Takeaways:

- The ROUTE feature dominates the prediction with a large positive effect
- Distance and airtime significantly reduce the final output
- Most other operational factors have minimal impact in this case

- The net effect of all features moves the prediction from the expected +1.39 to -11.21

Key Insights and Recommendations

1. **Seasonal Patterns:** Delays peak during summer and holiday seasons
2. **Airline Variability:** Some carriers consistently perform better than others
3. **Route Hotspots:** Certain routes have significantly higher delay probabilities
4. **Weather Impact:** While less frequent, weather delays cause the longest disruptions

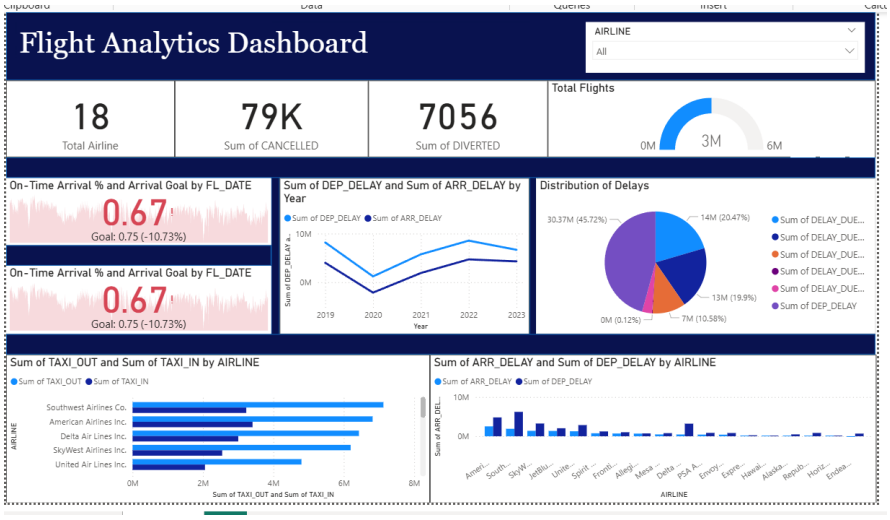
Recommendations for Airlines:

- Optimize scheduling during peak delay periods
- Improve turnaround processes for late aircraft scenarios
- Focus on high-risk routes with targeted improvements

Conclusion

This comprehensive analysis of flight delay data provides valuable insights for airlines, airports, and passengers. The predictive model demonstrates strong performance in forecasting delays, which could be integrated into operational systems to improve flight planning and customer communication. The visualizations effectively communicate patterns and relationships in the data, supporting data-driven decision making in the aviation industry.

We also created a dashboard for this project**.**



Dashboard Summary: Flight Delay Visualization

To enhance insights from our predictive analysis, we created an interactive Flight Analytics Dashboard using Power BI. This dashboard enables users to explore trends and key performance

indicators related to flight delays across different airlines and years.

Key Visual Elements:

- ## KPIs and Overview Tiles:
  - Total Airlines: 18
  - Cancelled Flights: Approximately 79,000
  - Diverted Flights: 7,056
  - Total Flights: Visualized with a gauge indicating ~3 million total records
- ## On-Time Arrival Rate:
  - The on-time arrival percentage stands at 0.67 compared to the goal of 0.75, indicating a -10.73% gap from the target performance.
- ## Yearly Delay Trends:
  - A line chart shows the sum of departure and arrival delays over the years 2019 to 2023.
  - Both delay types show a dip in 2020, followed by a gradual increase post-pandemic, peaking in 2022.
- ## Delay Distribution:
  - A pie chart visualizes delay causes:
    - The largest portion (45.72%) comes from carrier-related delays.
    - Other significant contributors include late aircraft and weather delays.
- ## Taxi Time Analysis:
  - A bar chart displays TAXI_OUT and TAXI_IN durations by airline.
  - Southwest Airlines has the highest taxi times, followed by American and Delta.
- ## Delay Comparison by Airline:
  - A grouped bar chart compares arrival and departure delays by airline.
  - Some airlines, such as American and SkyWest, show notably higher delay values.

Interactive Elements:

- The dashboard includes a slicer/filter for airline selection, enabling focused analysis on individual carriers.

This visualization provides a clear and interactive way to explore delay patterns, compare airline performance, and understand the root causes of disruptions in air travel.